8.2.1 Physical equilibrium

A system in which the amount of matter constituting different phases does not change with time is said to be in physical equilibrium. This involves no perceptible physical change in the system. To understand the physical equilibrium let us analyse the following phase changes.

Solid-liquid equilibrium

Let us consider the melting of ice in a closed container at 273 K. This system will reach a state of physical equilibrium in which the amount of water in the solid phase and liquid phase does not change with time. In the process the total number of water molecules leaving from and returning to the solid phase at any instant are equal.

If some ice-cubes and water are placed in a thermos flask (at 273K and 1 atm pressure), then there will be no change in the mass of ice and water.

At equilibrium,

Rate of melting of ice = Rate of freezing of water

$$H_2O\ (s) \rightleftharpoons H_2O\ (l)$$

The above equilibrium exists only at a particular temperature and pressure. The temperature at which the solid and liquid phases of a substance are at equilibrium is called the melting point or freezing point of that substance.

Liquid - Vapour equilibrium

Similarly, there exists an equilibrium between the liquid phase and the vapour phase of a substance. For example, liquid water is in equilibrium with its vapour at 373 K and1 atm pressure in a closed vessel.

$$H_2O\ (l) \rightleftharpoons H_2O\ (g)$$

Here

Rate of evaporation = Rate of condensation

The temperature at which the liquid and vapour phases of a substance are at equilibrium is called the boiling point and condensation point of that substance.

Solid - Vapour equilibrium

Consider a closed system in which the solid sublimes to vapour. In this process also, equilibrium can be established between these two phases. When solid iodine is placed in a closed transparent vessel, after sometime,the vessel gets filled up with violet vapour due to sublimation of iodine. Initially, the intensity of the violet colour increases, after sometime it decreases and finally it becomes constant, as the following equilibrium is attained.

$$I_2\ (s) \rightleftharpoons I_2(g)$$

More examples

$$\text{Camphor}\ (s) \rightleftharpoons \text{Camphor}\ (g)$$

$$NH_4Cl\ (s) \rightleftharpoons NH_4Cl\ (g)$$

8.2.2 Equilibrium involving dissolution of solids or gases in liquids

Solid in liquids

When you add sugar to water at a particular temperature, it dissolves to form sugar solution. If you continue to

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